

School Bus Tracking Software

User Roles & Permissions Document – Version 1

Purpose: Define every user type, responsibility and access boundary across the School Bus Tracking platform.

1. User Roles

Role	Primary Responsibility	Platform
Super Admin	Manage schools, system configuration, users, subscriptions and global settings.	Web
School Admin	Manage school operations, students, parents, routes, buses and announcements.	Web
Transport Manager	Manage drivers, buses, routes, daily assignments, trips and live tracking.	Web
Driver	Operate assigned daily trip and provide live GPS location.	Android App
Parent	Track linked child's bus, view ETA/status and receive notifications.	Android App
Student	Represents the student linked to a parent, route and stop. No independent operational access.	Parent/Admin

2. Super Admin

Purpose: Global platform owner responsible for system-wide administration.

Permissions:

- Create, activate, suspend and manage schools/tenants.
- Manage school-level administrators and global user roles.
- Configure global system settings.
- View system-wide operational statistics and reports.
- Manage subscription/licensing configuration if applicable.
- Access audit logs and security settings.

Restrictions:

- Should not modify operational data casually without audit logging.
- Critical actions should require confirmation and be recorded.

3. School Admin

Purpose: Primary administrator for one school and its transportation operations.

Permissions:

- Manage school profile and settings.
- Create and manage students.
- Create and manage parent accounts/links.
- Create routes and route stops.
- Register and manage buses.
- Register and manage drivers.
- Assign students to routes and stops.
- Create daily driver/bus/route assignments.
- View live bus locations for the school.

- Send school-wide and route-specific notifications.
- View trip history, delays and operational reports.

Restrictions:

- Access must be restricted to the assigned school.
- Cannot view or modify another school's students, parents, routes or buses.

4. Transport Manager

Purpose: Manage day-to-day school transportation operations.

Permissions:

- View all assigned buses and drivers.
- Create/update daily driver + bus + route + trip assignments.
- Monitor live buses.
- Start/stop/override trip status when operationally required.
- Manage delays and delay reasons.
- Handle breakdown/emergency updates.
- View route and trip history.
- Send route-specific notifications.

Restrictions:

- Should not change high-level school configuration unless explicitly permitted.
- Student master-data access should be limited to transportation needs.

5. Driver

Purpose: Operate the assigned trip and provide live location from the driver mobile app.

Permissions:

- Login using mobile number + OTP or approved authentication.
- View today's assigned bus, route and trip.
- Confirm assigned trip.
- Start trip.
- Share live GPS location while the trip is active.
- View route stops and trip status.
- Mark operational events where required.
- Report delay, breakdown or emergency.
- End trip.

Restrictions:

- Cannot select an arbitrary student.
- Cannot change the master route or student assignment.
- Cannot access another driver's trip unless reassigned by authorized staff.
- Location sharing should operate only according to trip/business rules.

6. Parent

Purpose: Track and receive transportation information for linked children.

Permissions:

- Login using mobile number + OTP.
- View linked children.
- Switch between children.
- View assigned route and stop.
- View current bus location when available.
- View trip status and ETA.
- Receive automatic bus notifications.
- Receive school-wide announcements applicable to the parent.
- Receive route-specific announcements applicable to the child's route.

Restrictions:

- Parent can only view children linked to the account.
- Parent cannot change route, stop, bus or driver assignment.
- Parent cannot view other students' personal information.
- Live location access should be limited to the relevant trip/route.

7. Student

Purpose: Represents the child receiving school transportation.

Permissions:

- Student record stores identity, class/section and transportation assignment.
- Student is linked to a parent/guardian.
- Student is assigned to a route and a specific stop.
- Student's transport status can be used by authorized school staff.

Restrictions:

- No independent administrative access is required in Version 1.
- Student data is visible only to authorized school staff and linked parent/guardian.

8. Permission Matrix

Feature	Super	School	Transport	Driver	Parent
Manage Schools	✓	—	—	—	—
Manage School Admins	✓	—	—	—	—
Manage Students	✓	✓	View	—	View linked
Manage Parents	✓	✓	View	—	Own
Manage Routes	✓	✓	✓	View	View assigned
Manage Stops	✓	✓	✓	View	View assigned
Manage Buses	✓	✓	✓	View assigned	View assigned
Manage Drivers	✓	✓	✓	Own profile	—
Daily Trip Assignment	✓	✓	✓	View	View
Live Bus Tracking	✓	✓	✓	Own trip	Linked child
Trip History	✓	✓	✓	Own trips	Linked child
Delay Management	✓	✓	✓	Report	View
Emergency Update	✓	✓	✓	Report	View
School Notification	✓	✓	—	—	Receive
Route Notification	✓	✓	✓	—	Receive
Global Settings	✓	Limited	—	—	—
Audit Logs	✓	School	Transport	Own activity	—

9. Core Access Rules

- Every operational user belongs to a school/tenant unless they are a global Super Admin.
- Role-based access control (RBAC) must be enforced on both frontend and backend.
- Frontend hiding is not sufficient; every protected API must validate role and school scope.
- Parents can access only their linked children.
- Drivers can access only their assigned/current trips.
- Transport staff can access transportation data within their authorized school.
- All important create/update/delete/assignment actions should be audit logged.
- Daily driver/bus/route assignment is separate from permanent master records.

10. Authentication

- Parent: Mobile Number + OTP.
- Driver: Mobile Number + OTP or secure staff authentication.
- School Admin / Transport Manager: Secure login with role-based access; OTP/2FA can be added.
- Super Admin: Strong authentication with mandatory 2FA recommended.
- Session/token expiry, logout, device/session control and account suspension should be supported.

11. Role Hierarchy

Super Admin → **School Admin** → **Transport Manager** → **Driver**

School Admin manages **Students + Parents + Routes + Buses + Drivers**.

Parent sees only linked **Students** → **Route** → **Stop** → **Bus Trip**.